

STATISTICAL ANALYSIS OF KEY HEALTH INDICATORS PREDICTING DIABETES : A COMPREHENSIVE STUDY

Abstract

Our study confirms that blood glucose levels and HbA1c are the most crucial indicators of diabetes, with HbA1c presenting the best model accuracy and correlation. Age also plays a role, diabetic people were on average 20.8 years older, but did not alter the predictive power of the biomarker. BMI was significant but not very predictive. These findings highlight the critical role blood glucose markers play in assessing diabetes risk, but they also highlight the need for more precise data and flexible models to fully understand the underlying patterns.

1- INTRODUCTION

Throughout life, medical professionals advise us to adopt specific lifestyles to minimize the risk of developing diabetes. These recommendations often highlight multiple factors that can influence this risk including hba1c levels, glucose levels, age and the body mass index. In our paper, we aim to explore which of these factors contributes the most to the likelihood of developing diabetes.

The dataset used in this study includes various variables related to factors that could have influenced diabetes diagnoses in several individuals, along with a column indicating whether an individual has been diagnosed with diabetes. The variables were narrowed down to four of the most common causes: HbA1c levels, glucose levels, age, and body mass index (BMI). Statistical analyses will be conducted to determine which of these factors has the strongest association with the risk of developing diabetes.

2- DATA ACQUISITION & EXPLORATION

The data was sourced from kaggle , it includes various records such as age , blood pressure , hemoglobin and cholesterol levels in the body etc. These variables are frequently used to assess a person's risk of developing diabetes. The data set acts as a sample representing 100,000 entries.

2.1 - SUMMARY STATISTICS

Summary statistics for key variables :

- Mean of age : 41.9
- Median of age: 43
- Standard deviation of age: 22.5
- Mean of BMI: 27.3
- Median of BMI: 27.3
- Standard deviation of BMI: 6.6
- Mean of glucose: 138
- Median of glucose: 140

- Standard deviation of glucose: 40.7
- Mean of hemoglobin levels: 5.5
- Median of hemoglobin levels: 5.8
- Standard deviation of hemoglobin levels: 1.1

2- ANALYSIS OF DATA FREQUENCIES

This part of the analysis focuses on building frequency distributions for key variables in the dataset. It includes determining the range, class width, class limits, and class boundaries. These components are used to create graphical representations such as histograms, which help us visualize how the data is distributed. We also assess the shape of each graph to identify whether the distribution is symmetrical, uniform, or skewed, offering valuable insights into the characteristics of the data.

3.1 RANGES :

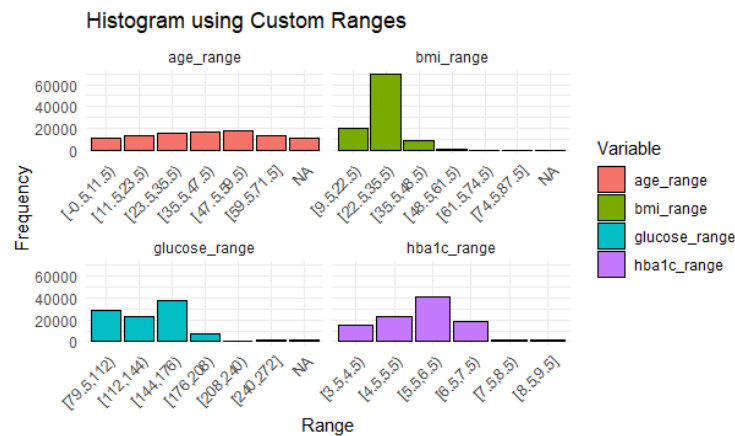


Figure 1. Histogram for the ranges

- Age range based on the dataset is semi uniform and unimodal.
- Bmi age graph based on the dataset is unimodal and right skewed.
- Blood Glucose range based on the dataset is bimodal.
- Hba1c levels range based on the dataset is right skewed.

3.2 CLASS WIDTHs :

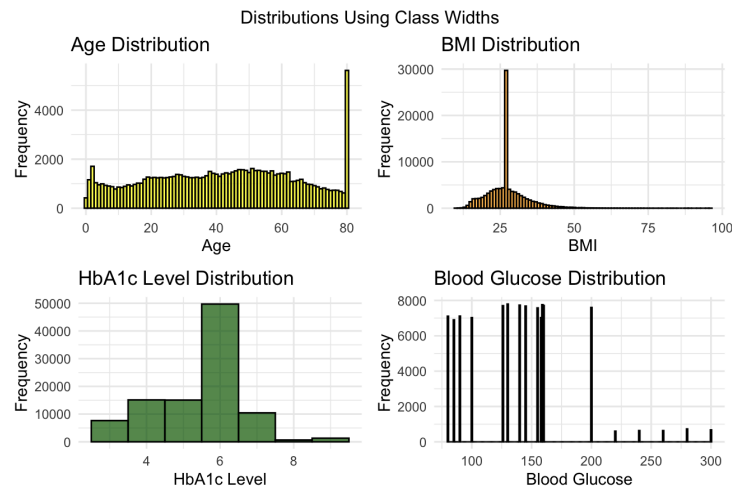


Figure 2. Histograms for the distributions using class widths

- Age distribution histogram is semi uniform with an outlier.
- Bmi distribution histogram is unimodal.
- Hba1c levels distribution histogram is unimodal.
- Blood glucose levels distribution graph is multimodal.

3.3 CLASS LIMITS :

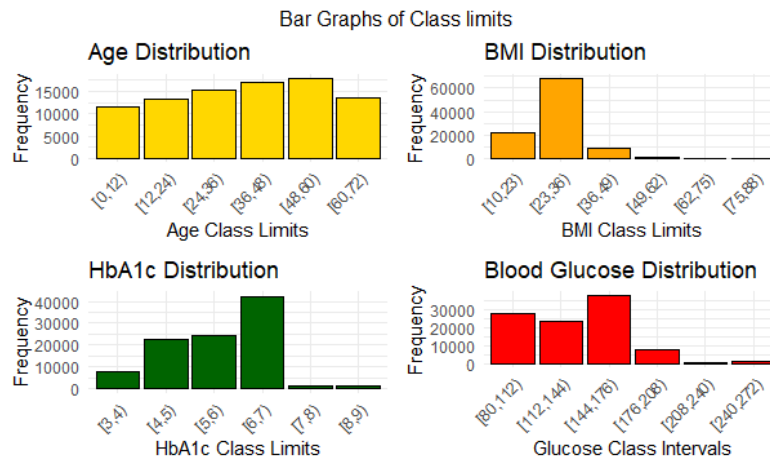


Figure 3. Bar graphs for class limits

- Age distribution is semi uniform
- Bmi distribution graph is unimodal and right skewed
- Hba1c distribution graph is left skewed
- Blood glucose distribution graph is multimodal and right skewed

3.4 CLASS BOUNDARIES :

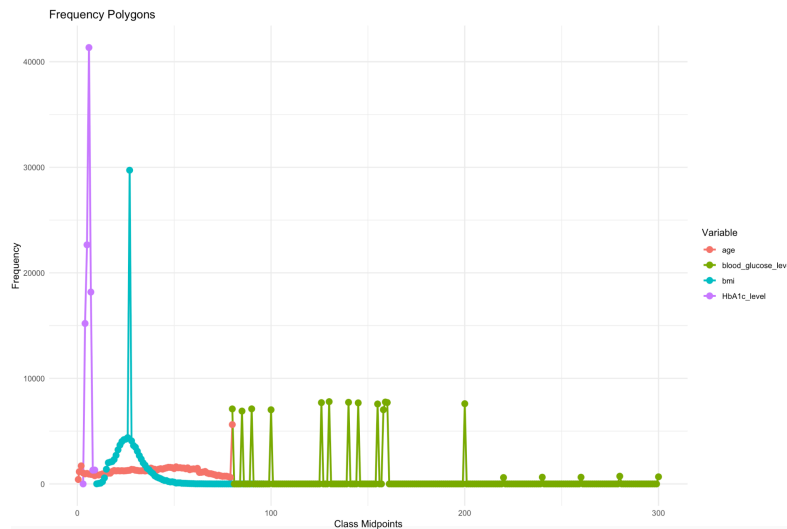


Figure 4. Frequency polygons for class midpoints

- Hba1c levels is the peak value
- Bmi levels is a second peak
- Blood glucose is uniform
- Age levels are left skewed

3.5 DISTRIBUTION OF KEY CONTINUOUS VARIABLES :

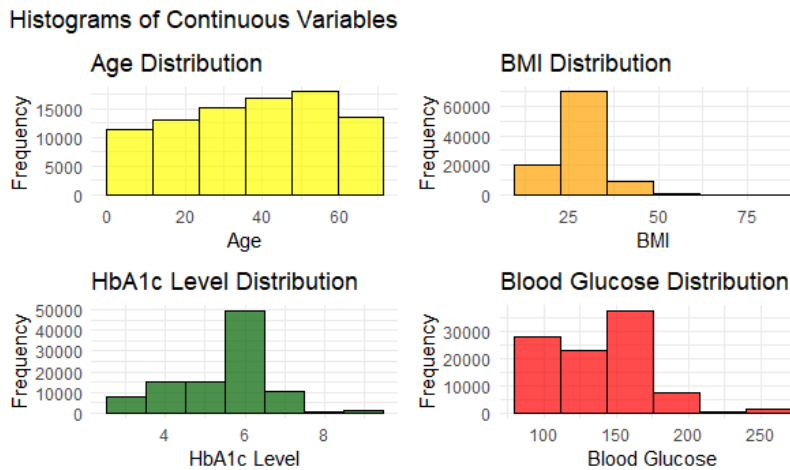


Figure 5. Histograms for the distributions of the key variables

- Age distribution graph is multimodal and left skewed
- Bmi distribution graph is unimodal and right skewed
- Hba1c level distribution is unimodal
- Blood glucose distribution graph is multimodal and right skewed

2.2 - VISUALIZATION OF KEY PREDICTORS IN RELATION TO DIABETES

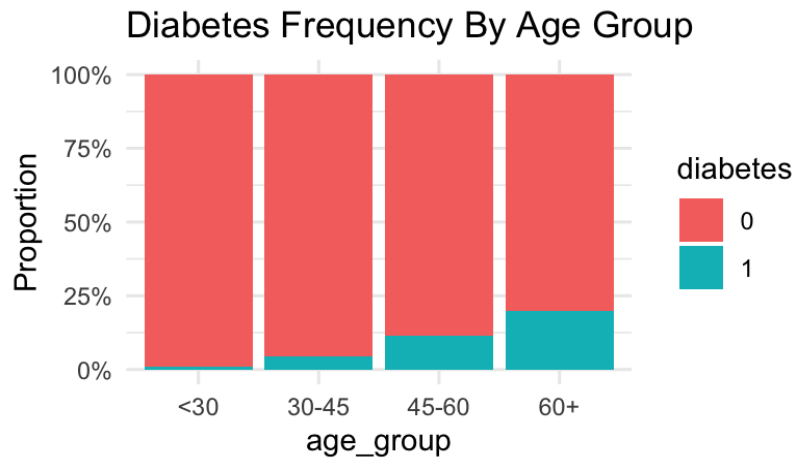


Figure 6. Barplot for frequency by age group

From this bar plot it's clear that as people age, the number of people with diabetes rises noticeably. Diabetes in the ages under 30 is relatively rare. Percentage of people with diabetes increases steadily showing that there's a positive correlation between age and diabetes occurrence in the dataset.

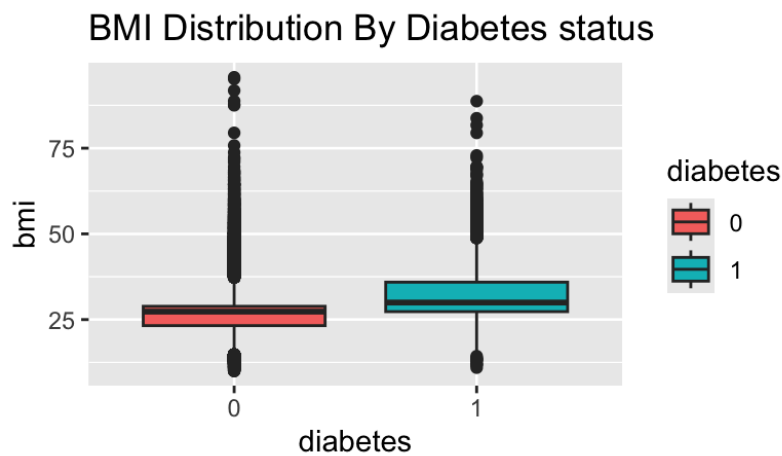


Figure 7. Boxplot for bmi distribution by diabetes status.

The box for people with diabetes seems to be shifted a little bit higher than the box for people without diabetes, suggesting that on average people with diabetes tend to have a higher BMI.

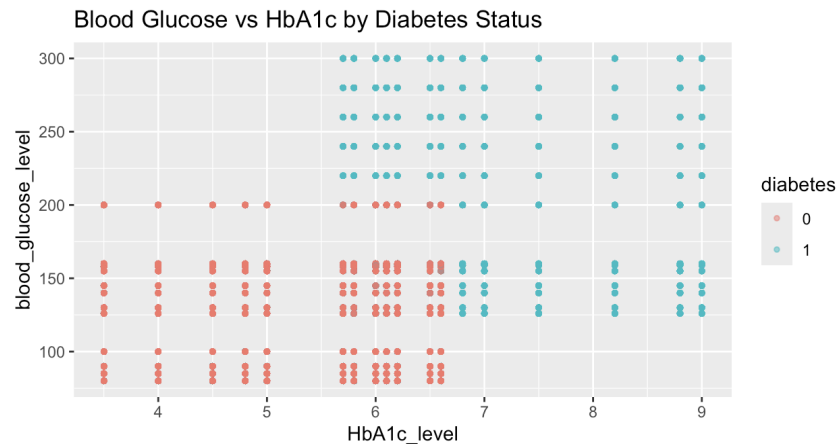


Figure 8. Scatter plot for Blood glucose vs Hb1Ac by diabetes status

The distribution of teal points mainly in the upper right section of the plot indicates that individuals with diabetes typically have higher blood glucose and HbA1c levels.

4- CENTRAL TENDENCY AND VARIABILITY ANALYSIS

4.1 MEASURES OF CENTRAL TENDENCY

	Mean	Median	Mode	Midrange
age	41.885856	43.00	80.00	40.04
bmi	27.320767	27.32	27.32	52.85
blood_glucose_level	138.058060	140.00	130.00	190.00
HbA1c_level	5.527507	5.80	6.60	6.25

Figure 9. Table for the measures of central tendencies

We summarized the distribution of key health indicators (age, bmi, blood glucose levels and HbA1c levels) using unweighted measures of central tendency. Weighted means were not used because the goal of this section was to describe the overall distribution of the data not adjust for sampling bias.

4.2 MEASURES OF VARIATION

	range	variance	std_var	coeff_of_var
age	79.92	507.008078	22.516840	0.5375762
bmi	85.68	44.046894	6.636783	0.2429208
blood_glucose_level	220.00	1657.152341	40.708136	0.2948624
HbA1c_level	5.50	1.146339	1.070672	0.1936989

Figure 10. Table for the measures of variation

We computed four main measures of variation. Age displayed the most variation. Significant variation was also seen in bmi and blood glucose levels, with glucose levels being especially dispersed. HbA1c levels on the other hand showed the least variation, indicating values were constant throughout the sample. The coefficient of variation expresses variation as a proportion of the mean, its used to compare the relative variability between variables with different units

4.3 MEASURES OF POSITION

	Z_Score_Above_2	Z_Score_Below_2	Percentiles	Outliers_Count
age	95675	0	24, 43, 60	0
bmi	100000	0	23.63, 27.32, 29.58	7086
blood_glucose_level	100000	0	100, 140, 159	2038
HbA1c_level	2619	0	4.8, 5.8, 6.2	1315

Figure 11. Table for the measures of position

We computed the measures of position for the key health indicators. The majority of the values for the age variable were grouped around the middle of the distribution with no extreme outliers in the data. On the other hand, BMI and blood glucose levels displayed a large number of outliers, indicating either there's significant variation or potentially extreme readings within these variables. In comparison to bmi and blood glucose levels, the HbA1c levels displayed a more concentrated distribution with fewer extreme values. These metrics help highlight variability of the data, emphasizing areas where outliers need further investigation.

5- TESTING FOR GAUSSIAN DISTRIBUTION

Q-Q plots were made in order to evaluate the normality of the key variables.

5.1 VISUALS

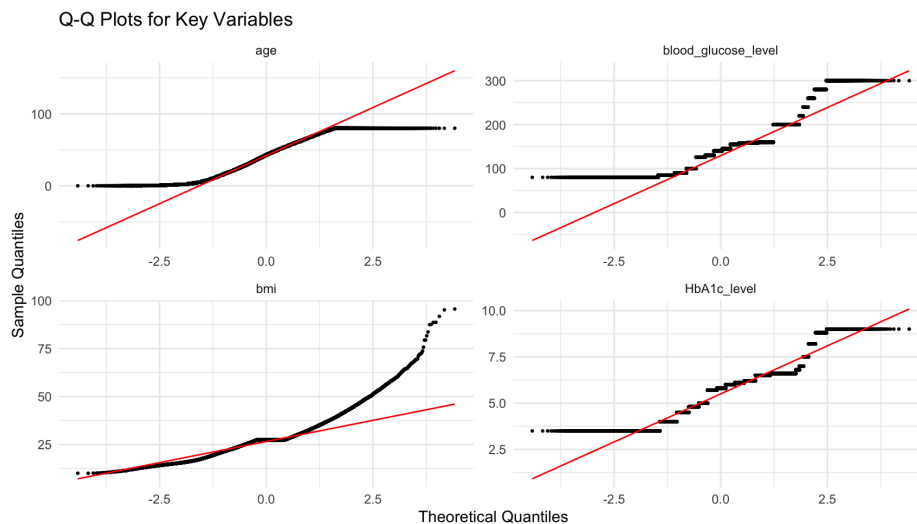


Figure 12. Normal Q-Q Plot

According to the Q-Q Plots, none of the key indicators follow a normal distribution. With evidence of skewness and division, all variables deviate from the theoretical normal line, especially blood glucose and Hb1Ac levels. The Shapiro-Wilk test further supports this conclusion. For BMI ($W = 0.9227$), HbA1c level ($W = 0.9337$), blood glucose level ($W = 0.9048$), and age ($W = 0.9677$) all returned p -values less than 0.05, indicating that none of the variables are normally distributed.

5.2 CENTRAL LIMIT THEOREM

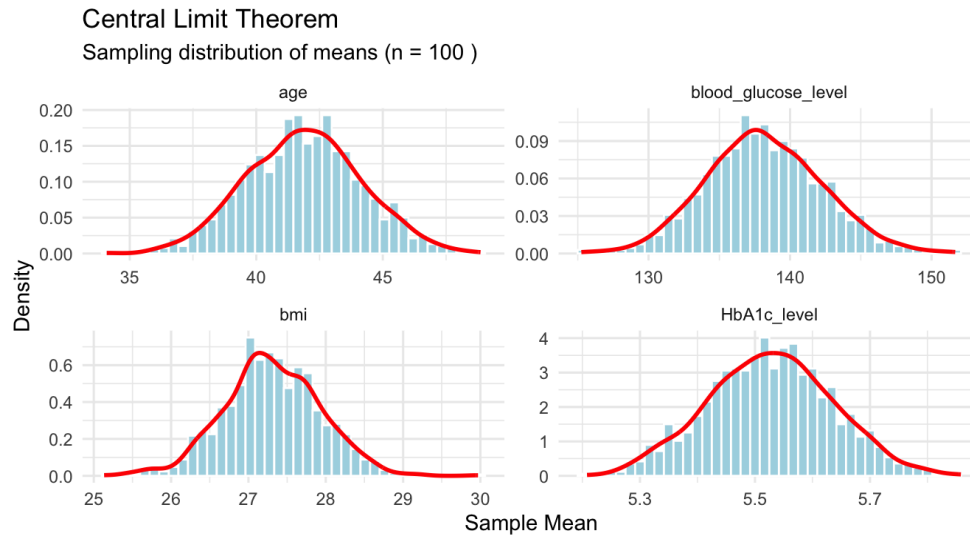


Figure 13. Central limit theorem demonstration with histograms

We tested the sample means with the Shapiro-Wilk normality test. A p-value greater than 0.05 indicates that the sample means are probably distributed.

- The Shapiro-Wilk test for age's sample means resulted in a p-value = 0.9649
- The Shapiro-Wilk test for bmi's sample means resulted in a p-value = 0.1767
- The Shapiro-Wilk test for blood glucose levels's sample means resulted in a p-value = 0.5944
- The Shapiro-Wilk test for HbA1c level's sample means resulted in a p-value = 0.6162

Since all the p-values are greater than 0.05, we fail to reject the Shapiro-Wilk test's null hypothesis which states that the sample means for age, bmi, blood glucose levels, and HbA1c levels are normally distributed, cannot be rejected because the p-values for each variable are greater than 0.05.

6- CONFIDENCE RANGE ANALYSIS

6.1 CONFIDENCE INTERVALS (T-TESTS)

- The mean age for people without diabetes is 40.12 years, while for those with diabetes is 60.95 years. The 95% confidence interval for the difference in means is between -21.17 and -20.49 years
- The mean BMI for people without diabetes is 26.89, while for those with diabetes is 31.99. The 95% confidence interval for the difference in means is between -5.27 and -4.94.
- The mean blood glucose for people without diabetes is 132.85, while for those with diabetes is 194.09. The 95% confidence intervals for the difference in means is between -62.51 and -59.98.
- The mean HbA1c level for people without diabetes is 5.40, while for those with diabetes is 6.93. The 95% confidence interval for the difference in mean is between -1.56 and -1.51.

7- INFERENCE ANALYSIS

To see how well independent variables predict the presence of diabetes. We ran a logistic regression. This method works well here because diabetes (dependent variable) is a binary outcome (either you have it or you don't) and the predictors (independent variables) like: age, BMI, glucose, and HbA1c are continuous.

HYPOTHESIS 1 : HBA1C IN RELATION TO DIABETES

H0 : HbA1c level has no significant effect on the likelihood of diabetes.

H1:HbA1c level has a significant effect on the likelihood of diabetes.

The logistic regression analysis evaluated how HbA1c level influences the probability of developing diabetes. The regression coefficient for HbA1c level was 2.4076 with a standard error of 0.0292. The z-value reached 82.47 which exceeded the critical value of +1.96,-1.96. The relationship between HbA1c level and diabetes shows statistical significance. The 95% confidence interval for the HbA1c level coefficient was [2.3509, 2.4653] and since it does not contain 0 we reject the null hypothesis. The p-value was $< 2e-16$ which is much smaller than the common alpha level of 0.05. Therefore, based on the p-value method we reject the null hypothesis.

The three methods of confidence interval, z-test and p-value all reject the null hypothesis. The analysis shows that HbA1c level is a significant predictor of diabetes in this dataset.

HYPOTHESIS 2 : AGE IN RELATION TO DIABETES

H0 : Age has no significant effect on the likelihood of diabetes.

H1:Age has a significant effect on the likelihood of diabetes.

The logistic regression analysis evaluated how Age influences the probability of developing diabetes. The regression coefficient for age was 0.0499 with a standard error of 0.00066. The z-value was 75.02 which exceeded the critical value of +1.96,-1.96. The relationship between age and diabetes shows statistical significance. The 95% confidence interval for the age coefficient was [0.0486, 0.0512] and since it does not contain 0 we reject the null hypothesis. The p-value was $< 2e-16$ which is much smaller than the common alpha level of 0.05. Therefore, based on the p-value method we reject the null hypothesis.

The three methods of confidence interval, z-test and p-value all reject the null hypothesis.. The analysis shows that age is a significant predictor of diabetes in this dataset.

HYPOTHESIS 3: BMI IN RELATION TO DIABETES

H0 : BMI has no significant effect on the likelihood of diabetes.

H1:BMI has a significant effect on the likelihood of diabetes.

The logistic regression analysis evaluated how BMI influences the probability of developing diabetes. The regression coefficient for BMI was 0.0941 with a standard error of 0.0015. The z-value was 62.90 which exceeded the critical value of +1.96,-1.96. The relationship between BMI and diabetes shows statistical significance. The 95% confidence interval for the BMI coefficient was [0.0912, 0.0970] and since it does not contain 0 we reject the null hypothesis. The p-value was $< 2e-16$ which is much smaller than the common alpha level of 0.05. Therefore, based on the p-value method we reject the null hypothesis.

The three methods of confidence interval, z-test and p-value all reject the null hypothesis. The analysis shows that BMI is a significant predictor of diabetes in this dataset.

HYPOTHESIS 4: GLUCOSE IN RELATION TO DIABETES

H0 : glucose level has no significant effect on the likelihood of diabetes.

H1:glucose level has a significant effect on the likelihood of diabetes.

The logistic regression analysis evaluated how Glucose Level influences the probability of developing diabetes. The regression coefficient for blood glucose level was 0.0343 with a standard error of 0.00034. The z-value was 100.65 which exceeded the critical value of +1.96,-1.96. The relationship between glucose level and diabetes shows statistical

significance. The 95% confidence interval for the glucose level coefficient was [0.0337, 0.0350] and since it does not contain 0 we reject the null hypothesis. The p-value was $< 2e-16$ which is much smaller than the common alpha level of 0.05. Therefore, based on the p-value method, we reject the null hypothesis.

The three methods of confidence interval, z-test and p-value all reject the null hypothesis. The analysis shows that blood glucose level is a significant predictor of diabetes in this dataset.

8- TESTING THE DIFFERENCE BETWEEN:

8.1 TWO MEANS

- AGE : individuals with diabetes (mean age = 60.95) were significantly older than those without diabetes (mean age = 40.1) , $t(12560) = -119.59$, $p < 0.001$, $d = -0.96$ (large effect). The 95% confidence interval indicates that the true mean age difference lies between 20.49 and 21.17
- BMI : Non diabetics (mean bmi = 26.9) have a lower bmi than diabetics do (mean bmi = 32.0), $t(9654.9) = -60.27$, $p < 0.001$, $d = -0.79$ (medium-to-large effect). The 95% confidence interval indicates that the true mean bmi difference lies between between 4.94 and 5.27.
- BLOOD GLUCOSE : Non diabetes (mean glucose = 132.8 mg/dL) have a significantly lower glucose than those with diabetes (mean glucose = 194.1), $t(9045.3) = -94.80$, $p < 0.001$, $d = -1.66$ (very large effect). The 95% confidence interval indicates that the true mean blood glucose level lies between between 59.98 and 62.51
- HbA1c LEVELS : Non diabetes (mean HbA1c = 5.4%) have a significantly lower hemoglobin levels than those with diabetes (mean HbA1c = 6.9%), $t(9828.7) = -127.01$, $p < 0.001$, $d = -1.57$ (very large effect). The 95% confidence interval indicates that the true mean HbA1c level lies between 1.51 and 1.56.

8.2 TWO PROPORTIONS

We compared the proportion of individuals with hypertension in two different groups (diabetic vs non diabetic), diabetics had a significantly higher level of hypertension (24.6%) than non diabetics (5.9%), with an absolute difference of 18.7%. This difference was very significant ($\chi^2 = 3910.7$, $p < 0.001$). Hypertension is 4.2x more common with people with diabetes (24.6/5.9). The true difference , with a 95% confidence interval is between 17.7% and 19.6%.

8.3 TWO VARIANCES

We are comparing variances of key health indicators between diabetics and non diabetic individuals. We use the F-test to test if the variance of both groups are equal. The ratio of the two groups' variances is represented by the F-test statistic, along with a p-value that indicates whether this ratio is statistically different from 1. And to give an estimate of the actual ratio of variance, a 95% confidence interval is included.

- AGE : Diabetics showed a greater age variability than non diabetics , with a variance ratio of 2.35, a 95% CI of [2.28, 2.42] and a $p < 0.001$.
- BMI : Diabetes showed a lower bmi than non diabetics , with a variance ratio of 0.71, a 95% CI of [0.69, 0.73] and a $p < 0.001$
- BLOOD GLUCOSE : Diabetics showed significantly less blood glucose levels than non diabetics, with a variance ratio of 0.34, a 95% CI of [0.33, 0.35] and a $p < 0.001$.
- HbA1c LEVELS: Diabetics showed significantly lower HbA1c levels than non diabetics , with a variance ratio of 0.81, a 95% CI of [0.79, 0.84] and a $p < 0.001$.

9- PREDICTIVE MODELING AND ASSOCIATION TESTING

9.1 CORRELATION

In this section we performed a Pearson correlation analysis to examine the relationship between the key variables and the diabetes diagnosis. These relationships' direction and strength are measured by the Pearson correlation coefficient (r), and it ranges from -1 to 1, the closer the number to 1 the stronger the positive correlation and the closer the number to -1 the stronger the negative correlation. This analysis helps identify which factors are more closely associated with diabetes occurrence.

- Age and diabetes : Correlation coefficient = 0.258, suggesting a positive linear relationship that is weak to moderate. This indicates that as one ages, the probability of receiving a diabetes diagnosis tends to rise as well. With a $p < 2.2e-16$, correlation is statistically meaningful. The 95% confidence interval for the actual correlation is between [0.2522-0.2638]. Even though age appears to be associated with diabetes status, it represents just a minor portion of the variation.
- Bmi and diabetes : Correlation coefficient = 0.214, suggesting a positive linear relationship that is weak to moderate. This means individuals with higher bmi are slightly more likely to have diabetes. The p -value $< 2.2e-16$ and the 95% confidence interval for the correlation is [0.2084, 0.2203], even though the relationship is statistically significant, the correlation is still low.
- Blood glucose Levels and diabetes: Correlation coefficient = 0.420, suggesting a moderate positive linear relationship. This means that as blood glucose levels increase, individuals are more likely to be diagnosed with diabetes. The p -value $< 2.2e-16$ shows that this relationship is statistically significant. The 95% confidence interval for the correlation is [0.4144, 0.4247]. Given the moderate strength of this correlation, blood glucose level is a strong indicator of the likelihood of diabetes diagnosis
- HbA1c levels and diabetes: Correlation coefficient = 0.401, suggesting a moderate positive linear relationship. This means that as HbA1c levels increase, individuals are more likely to be diagnosed with diabetes. The p -value $< 2.2e-16$ indicates that this relationship is statistically significant. The 95% confidence interval for the correlation is [0.3954, 0.4059], indicating a reliable moderate association. This relationship shows a moderate strength, meaning that while HbA1c is important, other factors may also contribute to diabetes diagnosis.

Summary : The analysis reveals that the most significant correlation between diabetes diagnosis and the blood glucose levels is 0.420, suggesting a moderately positive association. Following next are the HbA1c levels exhibiting a correlation of 0.401.

9.2 REGRESSION

9.2.1 REGRESSION GRAPHS

In this section we will fit logistic regression models instead of linear regression because our response variable is binary (diabetes), using health indications as predictors and examine the coefficients to assess the direction and intensity of their relationship with the diabetes diagnosis.

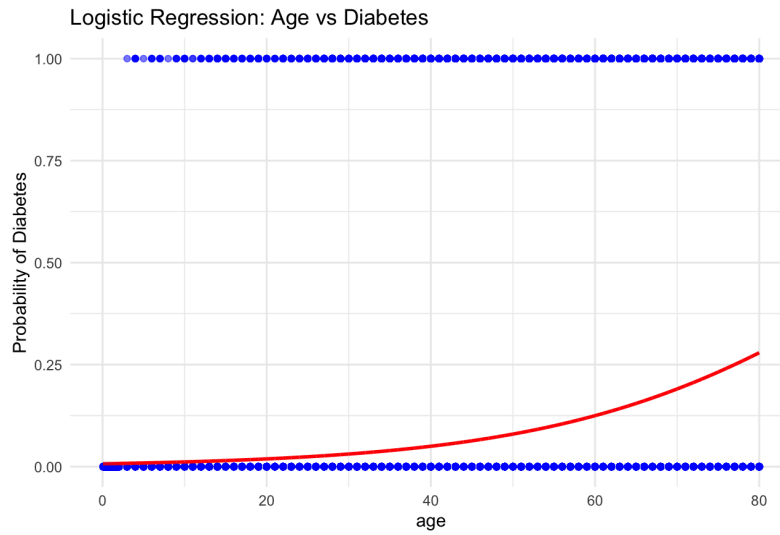


Figure 14. Scatter plot of age vs diabetes and regression line

The plot indicates that the likelihood of diabetes generally rises with age, the curve begins fairly level at younger ages and progressively rises as one gets older. This suggests that older people are more likely to develop diabetes.

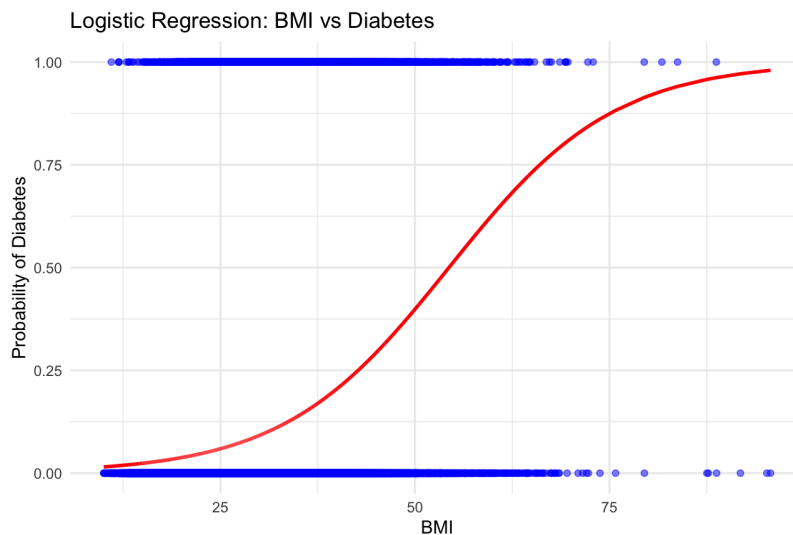


Figure 15. Scatter plot of bmi vs diabetes and regression line

As BMI rises, the likelihood of developing diabetes subsequently rises as well. The curve remains somewhat flat at lower BMI levels, signifying a minimal likelihood of diabetes. Yet, as BMI increases, the curve becomes sharper, indicating a quicker rise in the chance of developing diabetes. As BMI values increase, the curve begins to level off again, nearing a high chance of diabetes.

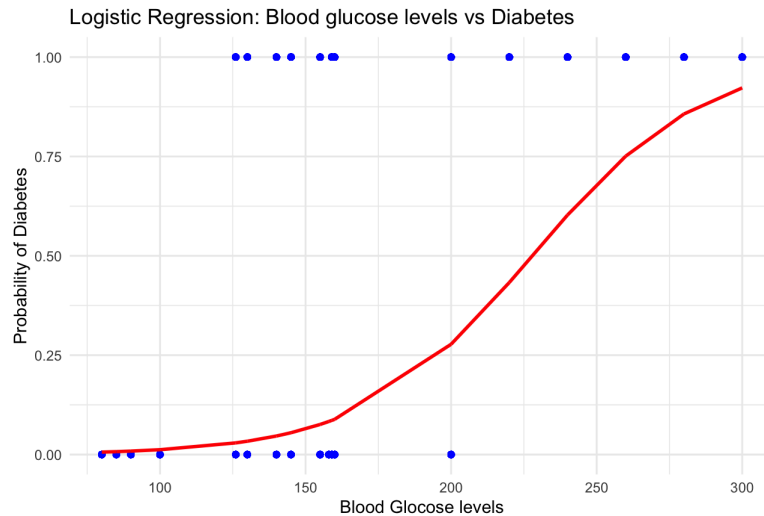


Figure 16. Scatter plot of blood glucose levels vs diabetes and regression line

Higher blood glucose levels correlate with an increased likelihood of diabetes. The plot shows a slow rise in the likelihood at low to moderate glucose levels, followed by a sharper rise at higher levels. This is consistent with the recognition that high blood sugar levels are a significant sign of diabetes.

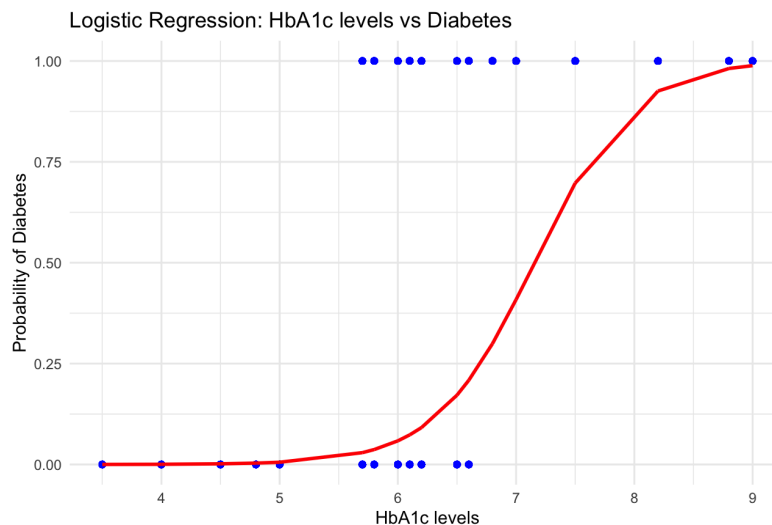


Figure 17. Scatter plot of HbA1c levels vs diabetes and regression line

There is a significant positive correlation between HbA1c levels and the likelihood of diabetes. The curve remains relatively flat at lower HbA1c levels, suggesting a minimal likelihood of diabetes. Nevertheless, past a specific threshold (approximately 6-7), the likelihood of diabetes rises steeply, attaining a significant probability at elevated HbA1c levels.

9.2.2 LOGISTIC REGRESSION DIAGNOSTICS

Since we did logistic regression, traditional metrics like R^2 (used in linear regression) aren't applicable here so we're going to be using pseudo- R^2 measures. Here are the logistic regression pseudo- R^2 values predicting diabetes from :

1- AGE : McFadden $R^2 = 0.126$, this suggests a slight fit, the model including age accounts for 12.6% of the variation in the probability of diabetes compared to a model without predictors. Nagelkerke R^2 (r^2_{CU}) = 0.160, this adjusted value

shows that 16% of the variations in diabetes status can be attributed to age, which is considered to have a moderate effect for logistic regression. In summary, age moderately strongly predicts diabetes status.

2- BMI : McFadden $R^2 = 0.067$, this is a weak fit. BMI alone accounts for only 6.7% of the variation in diabetes status. Nagelkerke R^2 (r2CU) = 0.087 , this adjusted value further shows a weak correlation, revealing that BMI accounts for 8.7% of the variability in diabetes results. In conclusion, bmi is a weaker predictor of diabetes compared to age.

3- HbA1c LEVELS : McFadden $R^2 = 0.34$, this is a moderate to strong model fit, indicating that HbA1c levels account for 34% of the variation in diabetes status. Nagelkerke R^2 (r2CU) = 0.41, this adjusted value says that HbA1c levels explains 34% of the variance in the diabetes outcome. In conclusion, HbA1c levels are the strongest predictors compared to age and bmi.

4- BLOOD GLUCOSE LEVELS : McFadden $R^2 = 0.26$, model explains 26% of the variation in the diabetes outcome, which is more than bmi and age but not HbA1c levels. Nagelkerke R^2 (r2CU) = 0.32 , 32% of the variation in the diabetes outcome is explained by glucose levels.

9.2.3 RESIDUAL PLOTS FOR MODEL DIAGNOSTICS

In this subsection, we plotted deviance residuals vs our predictors (age , bmi, blood glucose and the HbA1c levels) to visually see if our logistic regression model is appropriate. The deviance residuals are the difference between observed and predicted values on the log-likelihood scale. Ideally the residuals should be randomly scattered around zero with no visible pattern, this would mean that the model did a good job of capturing the relationships in the data. Seeing visible patterns or trends would indicate issues such as missing values, non-linearity in the data.

1. DEVIANCE RESIDUALS VS AGE :

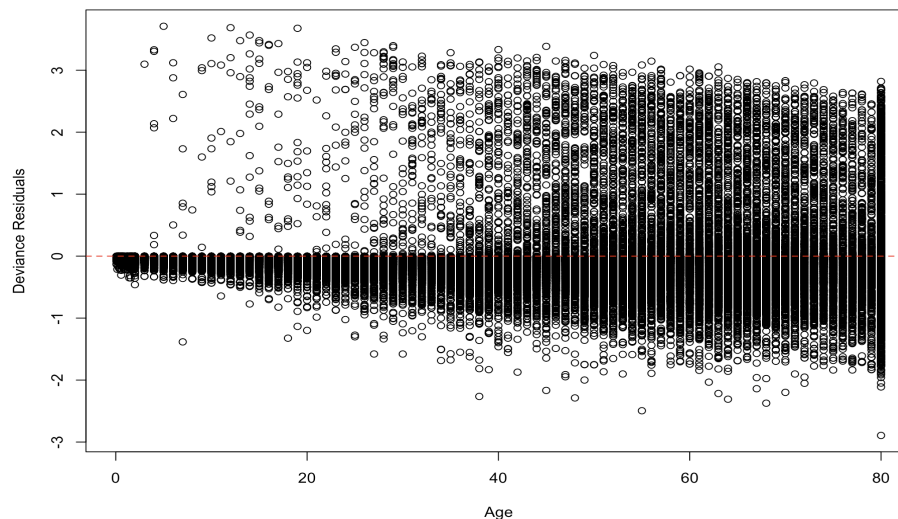


Figure 18. Scatter plot for deviance residuals and age.

The plot for deviance residuals against age indicates that our logistic regression model may not entirely capture a possible non-linear relationship. The residuals appear to be scattered around the zero at the younger and older ages which means that the middle age group seems to be having the greatest prediction error.

2. DEVIANCE RESIDUALS VS BMI :

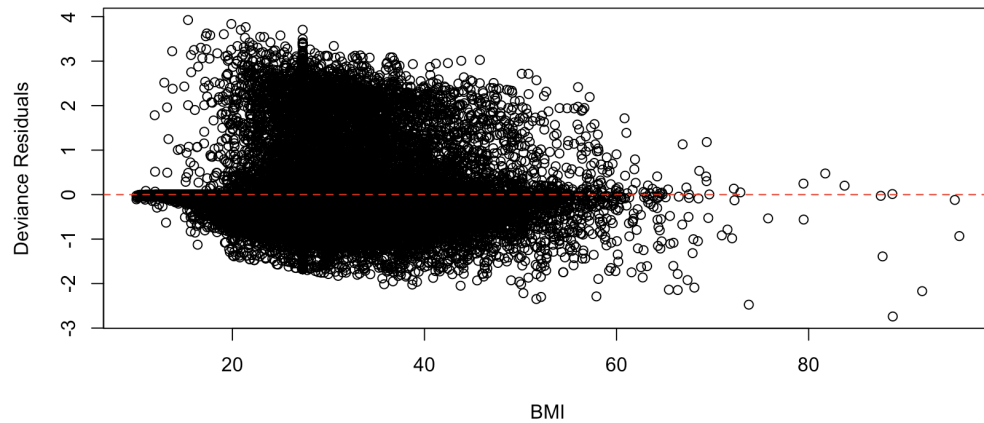


Figure 19. Plot for deviance residuals against BMI.

The deviance residuals plot against BMI shows that the model fits well for people with lower BMIs (less than 25), however, prediction errors rise and become more variable as BMI increases. This implies that the model might not be accurate for people with higher BMIs.

3. DEVIANCE RESIDUALS VS BLOOD GLUCOSE LEVELS :

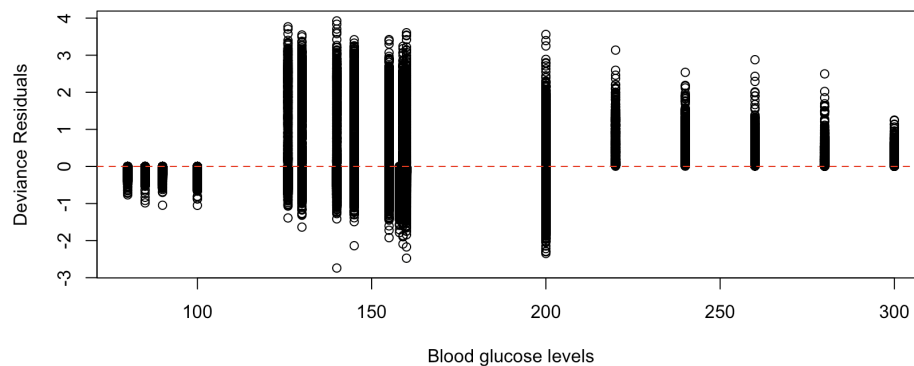


Figure 20. Plot for deviance residuals against blood glucose levels.

The deviance residual plot against blood glucose levels shows that the model performs well at lower glucose levels (around 80-100), with residuals clustered near zero. However, at higher glucose levels (around 150 and above), prediction errors increase and become more spread out. This suggests the model may struggle to predict diabetes outcomes for individuals with high blood glucose.

4. DEVIANCE RESIDUALS VS HbA1c LEVELS :

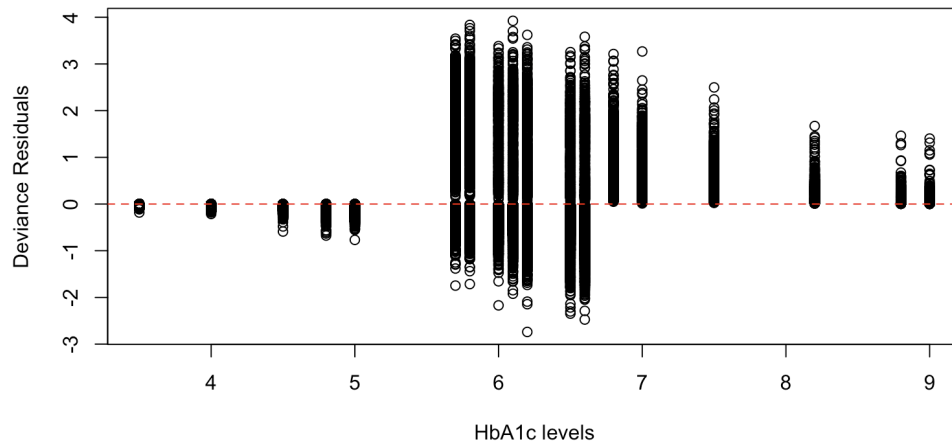


Figure 21. Plot for deviance residuals against HbA1c levels.

Similar to the blood glucose plot, residuals are clustered around zero at lower HbA1c levels (around 4-5), but as HbA1c levels rise (above 6), prediction errors increase significantly. This suggests that the model may not fully capture the relationship between high HbA1c and diabetes, indicating a need to explore non-linear effects involving HbA1c.

9.2.4 PREDICTED PROBABILITY PLOTS

In this part we visualized the predicted probabilities of diabetes based on the logistic regression model, focusing on how each predictor affects the likelihood of diabetes. The plots show how the probability varies with each predictor, including a 95% confidence interval highlighting the uncertainties with these predictions. Looking at these plots will help us evaluate the models effectiveness and gain more deeper insights on the connections between these predictors and diabetes.

1. PREDICTED PROBABILITY VS AGE :

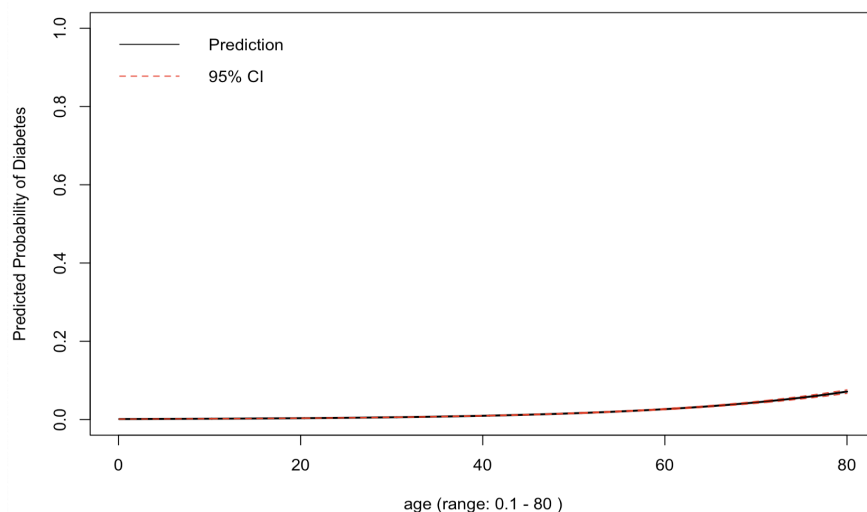


Figure 22. Plot for predicted probability against age.

This plot shows that there's a weak connection between the predicted probability and the age. The probability stays low for the majority of the age range then starts to slowly increase as age increases (60 to 80 years old). The confidence intervals are very narrow suggesting that there's a slight increase in the risk of diabetes as the age increases.

2. PREDICTED PROBABILITY VS BMI :

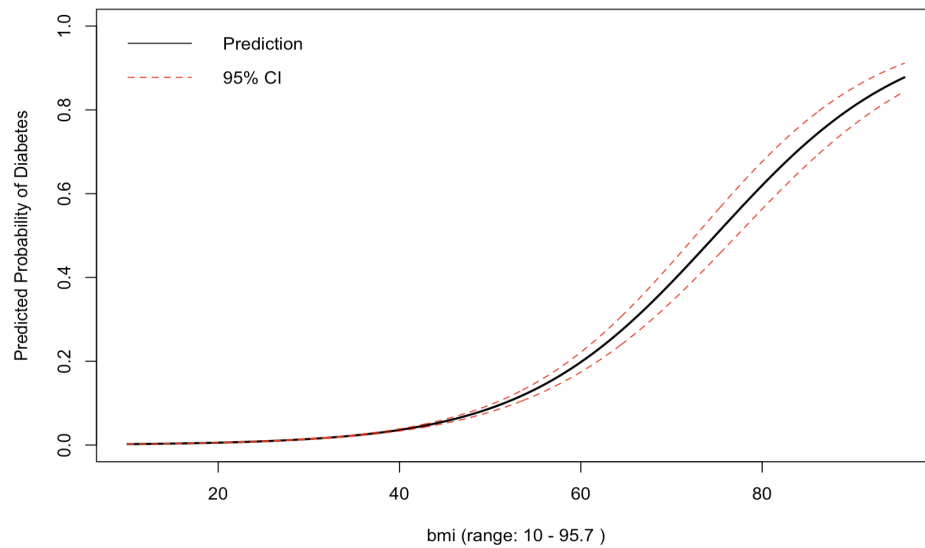


Figure 23. Predicted probability of diabetes against BMI.

The plot shows a significant connection between the predicted probability and the BMI. The probability starts low at lower BMI levels then starts to increase steadily especially after a BMI of 30 - 40. At the extreme BMIs, the confidence intervals widen implying that there's great uncertainty there. Overall, the model suggests that a higher BMI is clearly linked to a higher risk of diabetes.

3. PREDICTED PROBABILITY VS BLOOD GLUCOSE LEVELS :

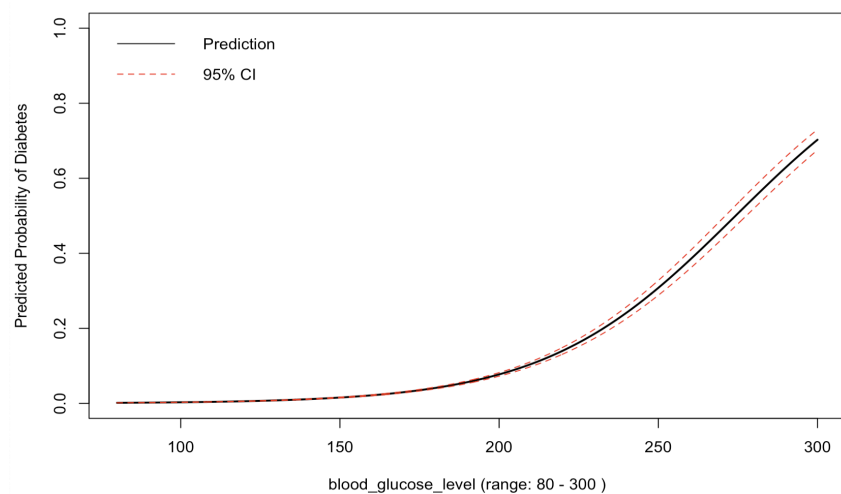


Figure 24. Predicted probability of diabetes against blood glucose levels.

The plot shows that the predicted probability of diabetes and the blood glucose levels are highly correlated. At lower blood glucose levels the probability is low, but then it increases rapidly after levels 150 - 200. The pattern is consistent, shown by the narrow confidence intervals. Overall, the model clearly shows that higher blood glucose levels strongly increase the likelihood of diabetes.

4. PREDICTED PROBABILITY VS HbA1c LEVELS :

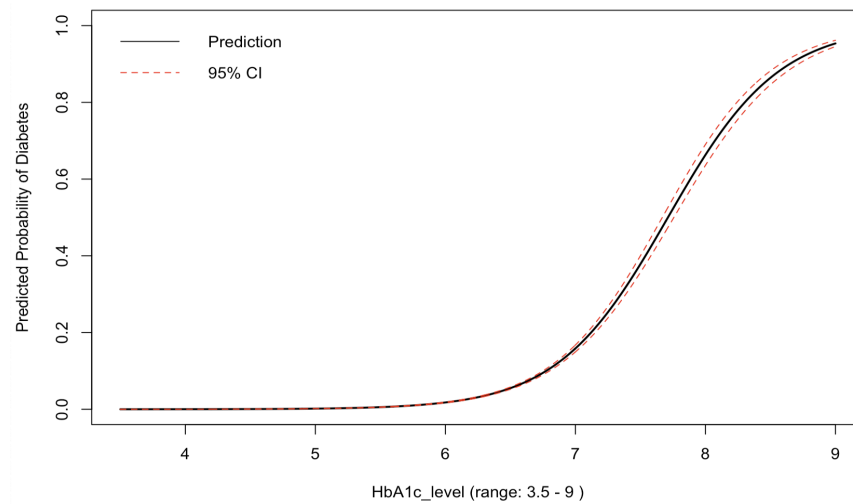


Figure 25. Predicted probability of diabetes against HbA1c levels.

The plot indicates a strong correlation between the predicted probability of diabetes and the HbA1c level. The probability is low at low HbA1c levels but increases significantly as HbA1c rises, especially after 6–7. Strong certainty in this trend is suggested by the narrow confidence intervals. In this model, HbA1c is a good indicator of diabetes risk overall.

10- GOODNESS-OF-FIT TEST

10.1 GOODNESS OF FIT TEST

The observed distribution of diabetes status was compared to an expected equal distribution using the Chi-squared Goodness Of Fit test, the findings indicated a significant difference ($X^2(df) = 68890$, $p < 0.001$), showing that the percentage of people without diabetes is (91.5%) is much higher than what would be predicted from an equal distribution.

10.2 TEST FOR INDEPENDENCE

1- Between hypertension and diabetes : the relationship between diabetes and hypertension was examined using the Chi-squared test of independence. The results indicated a statistically significant relationship between the two. We used Cramér's V rule to test the strength of this relationship and the result of this test was 0.198, indicating a small to medium effect size meaning that individuals with diabetes are more prone to hypertension than those without diabetes.

2- Between gender and diabetes : the relationship between gender and diabetes was examined using the Chi-squared test of independence. The results indicated a statistically significant relationship between the two. However, Cramér's V = 0.038 showed that this relationship was extremely weak. This implies that although diabetes and gender are related, the prevalence of diabetes in this group is not significantly influenced by gender.

10.3 TEST FOR HOMOGENEITY

To determine whether the distributions of categorical variables (hypertension and heart disease) in diabetics and non-diabetics are homogeneous. Additionally, we conducted the Pearson chi squared test with Yates continuity correction here.

1- Between hypertension and diabetes: The findings showed a significant difference ($X^2 = 3910.7$, $p < 2.2e-16$), indicating a strong correlation between the presence of hypertension and diabetes. This suggests that the two groups' distributions of hypertension are not uniform and differ greatly from one another.

2- Between heart disease and diabetes : The test result showed a significant difference ($X^2 = 2945.8$, $p < 2.2e-16$), indicating that these two groups have different distributions of heart disease. This suggests that diabetes and heart disease are closely related. With a Cramér's V of 0.17, this relationship has a moderate strength and a small to moderate effect size.

10.4 TEST FOR NORMALITY

The Kolmogorov-Smirnov test indicates that none of the key variables (age, bmi, blood glucose, HbA1c levels) follow a normal distribution.

11- VARIANCE ANALYSIS (ANOVA)

In this phase, we use ANOVA tests on our dataset to evaluate the mean differences of different groups.

According to the Q-Q plots, none of the key indicators :age, BMI, blood glucose level and HbA1c level follow a normal distribution. Additionally, the null hypothesis(H_0) of the Shapiro-Wilk test that assumes the normality of the sample means cannot be rejected since each variable has a p-value greater than 0.05.

And since diabetes is a binary variable we treated it as the independent variable in the ANOVA tests and key predictors (age, BMI, HbA1c level and blood glucose level) as dependent variables. Despite the lack of normality,we proceeded with ANOVA analysis to see if any significant mean differences of these variables exist between those with diabetes and those without.

11.1 ONE -WAY ANOVA RESULTS:

1 - AGE GROUP:

H_0 (Null Hypothesis):Age does not play a distinguishing role between individuals with and without diabetes.

H_1 (Alternative Hypothesis): Age does play a distinguishing role between individuals with and without diabetes.

The one-way ANOVA shows a highly significant difference in mean age between individuals with and without diabetes ($p\text{-value} < 2e-16$) suggesting that age distribution differs between the two groups. Tukey's HSD test confirms this with a mean difference of 20.83 years ($p\text{-value} = 0$).

```
> summary(anova_age)
              Df    Sum Sq Mean Sq F value Pr(>F)
as.factor(diabetes)    1  3375025  3375025    7131 <2e-16 ***
Residuals           99998  47325276     473
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure 26. Summary of anova on age.

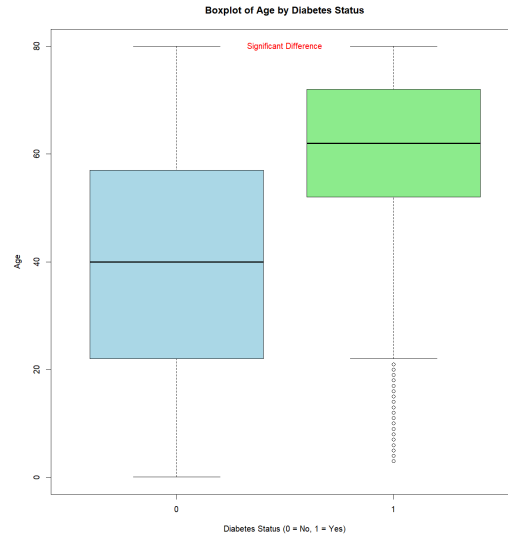


Figure 27. Boxplot of Age by Diabetes Status

In conclusion, there is a statistically significant difference in age between individuals with and without diabetes.

2- GLUCOSE GROUP:

H0 (Null Hypothesis): Glucose level does not play a distinguishing role between individuals with and without diabetes.

H1 (Alternative Hypothesis): Glucose level does play a distinguishing role between individuals with and without diabetes..

The One-Way ANOVA for blood glucose level indicates a highly significant difference between individuals with and without diabetes ($p\text{-value} < 2e-16$) suggesting that diabetes status significantly affects blood glucose levels. Tukey's HSD test further confirms this with a mean difference of 61.24 mg/dL between the two groups ($p\text{-value} = 0$).

```
> summary(anova_glucose)
              Df    Sum Sq Mean Sq F value Pr(>F)
as.factor(diabetes)  1 29170381 29170381  21363 <2e-16 ***
Residuals        99998 136543196    1365
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure 28.

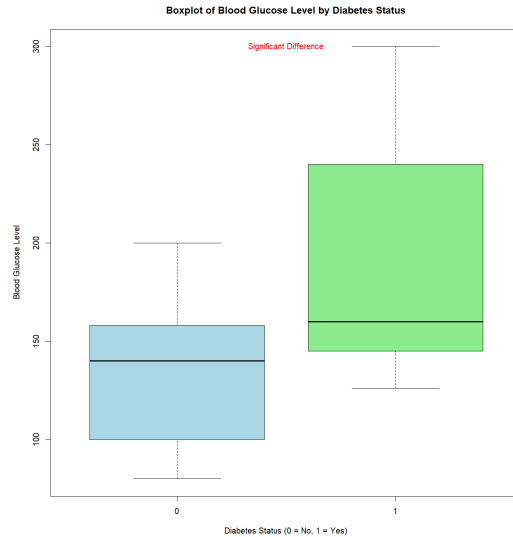


Figure 29. Boxplot of Blood Glucose Level by Diabetes Status

In conclusion, blood glucose levels are significantly higher in individuals with diabetes compared to those without diabetes.

3 - HBA1C GROUP:

H0 (Null Hypothesis): HBA1C level does not play a distinguishing role between individuals with and without diabetes.

H1(Alternative Hypothesis): HBA1C level does play a distinguishing role between individuals with and without diabetes.

The One-Way ANOVA for HbA1c levels shows a highly significant difference between individuals with and without diabetes (p-value < 2e-16) suggesting that diabetes status strongly influences HbA1c levels. Tukey's HSD test confirms this finding revealing a mean difference of 1.54 between the two groups (p-value = 0).

```
> anova_hba1c <- aov(HbA1c_level ~ as.factor(diabetes), data = data)
> summary(anova_hba1c)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
as.factor(diabetes)	1	18402	18402	19122	<2e-16 ***
Residuals	99998	96231	1		

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
> |
```

Figure 30.

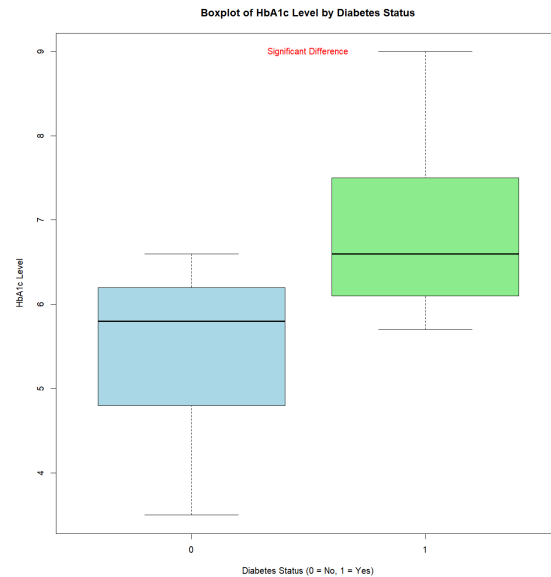


Figure 31. Boxplot of HbA1c Level by Diabetes Status

In conclusion, HbA1c levels are significantly higher in individuals with diabetes compared to those without diabetes.

4- BMI GROUP:

H0 (Null Hypothesis): BMI does not play a distinguishing role between individuals with and without diabetes.

H1 (Alternative Hypothesis): BMI does play a distinguishing role between individuals with and without diabetes.

The One-Way ANOVA for BMI shows a significant difference between individuals with and without diabetes (p-value < 2e-16) suggesting that diabetes status has a strong effect on BMI. Tukey's HSD test further supports this with a mean difference of 5.10 between the two groups (p-value = 0).

```
> summary(anova_bmi)
```

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
as.factor(diabetes)	1	202389	202389	4816	<2e-16 ***
Residuals	99998	4202256	42		

 Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Figure 32. Summary of anova on bmi.

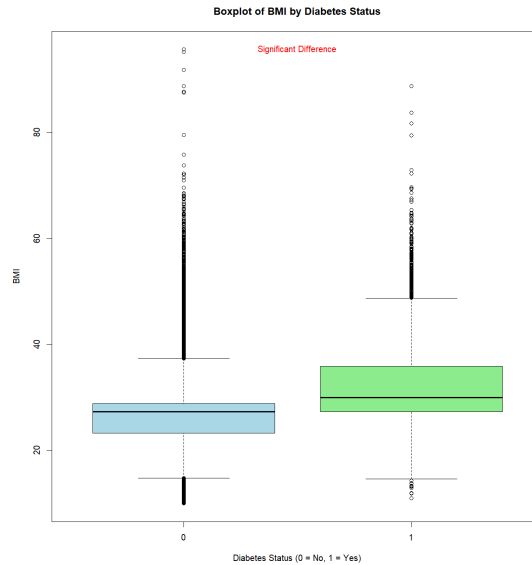


Figure 33. Boxplot of BMI by Diabetes Status

In conclusion, individuals with diabetes have a significantly higher BMI compared to those without diabetes.

11.2 TWO-WAY ANOVA RESULTS

ANOVA Analysis of HbA1c Levels by Diabetes Status and Age Group:

The Two-Way ANOVA results show that diabetes significantly affects HbA1c levels ($p\text{-value} < 2e-16$) while age group ($p\text{-value} = 0.485$) and the interaction ($p = 0.937$) do not. This basically means that diabetes influences HbA1c levels regardless of age group. Tukey's test also supports this, showing a clear difference between people with and without diabetes ($p\text{-value} = 0$) but no significant differences between age groups (all $p\text{-values} > 0.05$). Significant interaction pairs only occur when comparing diabetic vs. non-diabetic individuals within the same age group.

```
> summary(anova_hba1c)
              Df Sum Sq Mean Sq  F value Pr(>F)
diabetes         1  18402    18402  19121.748 <2e-16 ***
age_group        2      1      1      0.724  0.485
diabetes:age_group 2      0      0      0.065  0.937
Residuals      99994  96229      1
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure 34. ANOVA of HbA1c by Diabetes Status and Age Group.

In conclusion, diabetes has a significant effect on HbA1c levels while age and the interaction between diabetes and age group do not significantly affect HbA1c levels.

ANOVA Analysis of Blood Glucose Levels by Diabetes Status and Age Group:

The Two-Way ANOVA results show that diabetes significantly affects blood glucose levels ($p < 2e-16$) while age group ($p = 0.219$) and the interaction ($p = 0.139$) do not. This basically means that diabetes influences blood glucose levels regardless of age group. Tukey's test also supports this, showing a clear difference between people with and without diabetes

(p-value = 0) but there is no significant differences between age groups (all p-values > 0.05). Significant interaction pairs only occurs when comparing diabetic vs. non-diabetic individuals within the same age group

```
> summary(anova_glucose)
              Df    Sum Sq Mean Sq  F value Pr(>F)
diabetes       1  29170381 29170381 21363.691 <2e-16 ***
age_group      2     4148    2074     1.519  0.219
diabetes:age_group  2     5383    2691     1.971  0.139
Residuals    99994 136533666    1365
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure 35. ANOVA of Blood Glucose by Diabetes Status and Age Group.

In conclusion, diabetes significantly impacts blood glucose levels while age and the interaction between diabetes and age group do not.

ANOVA Analysis of BMI by Diabetes Status and Age Group:

The Two-Way ANOVA results show that both diabetes (p-value < 2e-16) and age group (p-value < 2e-16) significantly affect BMI. Additionally, The interaction between them is also significant (p < 2e-16), meaning diabetes impacts BMI differently across age groups. Tukey's test confirms this, showing clear BMI differences between people with and without diabetes (p-value = 0) and between all age groups (Young vs Middle-aged vs Old, all p-value = 0). The interaction results show that many diabetes-age combinations matter.

```
> summary(anova_bmi)
              Df    Sum Sq Mean Sq F value Pr(>F)
diabetes       1  202389   202389 5434.93 <2e-16 ***
age_group      2  471236   235618 6327.23 <2e-16 ***
diabetes:age_group  2    7377    3689   99.06 <2e-16 ***
Residuals    99994 3723643     37
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Figure 36. ANOVA of BMI by Diabetes Status and Age Group.

In conclusion, both diabetes and age group significantly affect BMI with their interaction also playing an important role in determining BMI differences.

12- NON-PARAMETRIC APPROACHES TO DATA ANALYSIS

12.1 WILCOXON RANK-SUM TEST (MANN-WHITNEY U TEST)

We compared the distributions of each variable between people with diabetes (1) and without diabetes (0) using two-sided tests.

- HbA1c level : Participants with diabetes had substantially higher HbA1c values than those without ($U = 1.25 \times 10^8$, $p < 0.0001$), with a large effect size ($r = 0.329$), making it the strongest discriminator between groups.
- Blood glucose levels : Blood glucose was the next most powerful marker: those with diabetes showed clearly elevated readings compared to non-diabetics ($U = 1.69 \times 10^8$, $p < 0.0001$), with a moderate effect size ($r = 0.274$).
- Age : On average, the diabetes group was older than the non-diabetes group ($U = 1.78 \times 10^8$, $p < 0.0001$), though age had a smaller impact ($r = 0.262$) than the direct blood sugar measures.

- BMI : While BMI also varied ($U = 2.29 \times 10^8$, $p < 0.0001$), it showed the weakest relation ($r = 0.200$), indicating that body mass index plays a more indirect role when it comes to diabetes risk.

Summary : All four variables differed significantly between individuals with and without diabetes ($p < 0.0001$). HbA1c emerged as the most distinguishing factor, followed by blood glucose, then age, and lastly BMI.

12.2 - SPEARMAN RANK CORRELATION COEFFICIENT

We examined how diabetes status moves with each measure using Spearman's rank correlation, the results are as follows:

- Diabetes vs. HbA1c level : $\rho = 0.329$, as HbA1c increases, the chances of having diabetes increase, and this relationship is unlikely to be due to chance ($p < 2 \times 10^{-16}$).
- Diabetes vs. blood glucose level: $\rho = 0.274$, higher blood sugar readings increase simultaneously with diabetes, again with $p < 2 \times 10^{-16}$.
- Diabetes vs. age: $\rho = 0.262$, older age is modestly but very significantly associated with diabetes ($p < 2 \times 10^{-16}$).
- Diabetes vs. BMI: $\rho = 0.200$, greater body-mass index also nudges up diabetes risk, with the same low p-value ($p < 2 \times 10^{-16}$).

In each case, the positive ρ tells us that as one measure rises, so does the likelihood of diabetes but none is a perfect predictor on its own.

12.3 - WALD-WOLFOWITZ RUNS TEST

We evaluated whether the relative ordering of each variable versus diabetes status exhibited any non-random structure by applying the Wald–Wolfowitz runs test. For each variable (age, BMI, HbA1c level, and blood glucose level), values were first dichotomized at the median and then ordered by the diabetes indicator (0 = no diabetes, 1 = diabetes).

H_0 : The sequence of “high” (1) and “low” (0) observations is purely random (no association).

H_1 : The sequence is non-random, indicating an association between the variable and diabetes status.

- RUNS TEST BETWEEN AGE AND DIABETES: Z-statistic: -1.727 , p-value: 0.0842. Conclusion: We fail to reject H_0 . Diabetes status appears randomly distributed over age (no evidence of association).
- RUNS TEST BETWEEN BMI AND DIABETES: Z-statistic: -1.297 , p-value: 0.1948. Conclusion: We fail to reject H_0 . Diabetes status appears randomly distributed over BMI (no evidence of association).
- RUNS TEST BETWEEN HBA1C LEVEL AND DIABETES: Z-statistic: -12.896 , p-value: 0.0000. Conclusion: We reject H_0 . Diabetes status is non-randomly distributed over HbA1c levels (strong evidence of association).
- Runs Test between blood glucose level and diabetes: Z-statistic: -10.537 , p-value: 0.0000 . Conclusion: We reject H_0 . Diabetes status is non-randomly distributed over blood glucose levels (strong evidence of association)

13. RESULTS & INTERPRETATION

13.1 MEANING OF THE FINDINGS:

After thoroughly investigating the dataset, our findings show that HbA1c levels and blood glucose levels are the leading indicators of diabetes risk. HbA1c levels ($p = 0.329$) and blood glucose ($p = 0.274$) not only showed the strongest correlations but also delivered the highest predictive power in our logistic models (McFadden R^2 of 0.34 and 0.26, respectively). Clinically, this reaffirms that monitoring these biomarkers provides the most direct insight into diabetic pathology.

Age also played a meaningful role: on average, participants with diabetes were about 20.8 years older ($p < 0.001$). Though age correlated moderately ($p = 0.262$), its effect did not vary across age groups—HbA1c’s predictive strength remained consistent, suggesting that blood sugar control is equally critical regardless of age bracket.

In contrast, BMI emerged as a secondary factor. While diabetic individuals did have higher BMI (mean difference = 5.1; $p < 0.001$), its lower pseudo- R^2 (0.067) indicates that excess weight alone captures only a small slice of the overall risk profile.

13.2 SUPPORT FOR HYPOTHESIS:

- **HbA1c and Glucose:** The data leaves no doubt that both biomarkers showed very strong, statistically significant effects ($p < 0.001$), and their confidence intervals stayed well clear of zero. This confirms that changes in blood glucose readings directly track with diabetes risk.
- **Age:** Age also emerged as a clear predictor ($p < 0.001$). People in our sample with diabetes were older on average, though we saw no evidence that age altered how HbA1c or glucose levels predict risk. In other words, high blood glucose readings matter just as much for younger and older individuals alike.
- **BMI:** Body mass index was significant, too ($p < 0.001$), but its overall influence was noticeably smaller compared to the two primary blood glucose markers. Elevated BMI does signal increased risk, but it does not carry the same predictive weight as direct glucose measurement.

13.3 LIMITATIONS AND CONSIDERATIONS:

While we’ve uncovered clear patterns, this study has its boundaries. We captured a snapshot in time, so we can talk about who shows which traits, but we can’t say that one factor directly makes diabetes happen. Our data didn’t include everyday habits or family history—things like what people eat, how much they move, or genetic background—which almost certainly influence risk. We also simplified diabetes into a yes/no outcome, potentially glossing over early warning signs like prediabetes. Finally, to make sense of data that didn’t fit a neat bell curve, we leaned on non-parametric tests and logistic regression; these approaches come with assumptions that might miss twists and turns in the data. Future work using more adaptable models could help fill in those gaps.

Adding in lifestyle snapshots (diet, exercise), genetic insights, and social factors would enrich our understanding. Employing flexible, data driven tools like machine learning might uncover hidden relationships that traditional methods overlook. And digging into subgroups by gender, ethnicity, or diabetes subtype could guide truly personalized prevention and treatment strategies.

14. CONCLUSION

14.1 KEY TAKEAWAYS:

Putting HbA1c levels, blood glucose, age, and BMI side by side makes one thing clear: blood glucose readings are the most reliable warning signals for diabetes. Older age does add to the risk people with diabetes in our sample were generally older but it isn’t as telling as those glucose measurements. BMI also plays a part, yet it only scratches the surface of risk compared to the other indicators.

14.2 IMPLICATIONS FOR PRACTICE :

What does this mean for the real world? First, regular HbA1c and glucose checks belong at the top of any screening protocol, whether in a doctor's office or out in the community. Second, public health efforts should keep these markers front and center, while still reaching out to older adults and those with higher BMI to encourage testing.

14.3 LOOKING AHEAD :

If we had the chance to follow people over months or years, we'd gain a richer sense of how risk factors evolve—revealing who moves from healthy to prediabetic to diabetic, and why. Adding in lifestyle snapshots (like diet and exercise), DNA clues, and social factors would bring even more detail. Using flexible, data-driven tools, like machine learning, could shine a light on hidden patterns that escape traditional methods. And finally, digging into subgroups by things like gender, race, or specific diabetes types could point the way toward personalized prevention and care.

Overall, this work highlights why monitoring blood sugar has to be at the heart of diabetes prevention, while setting the stage for more holistic research that weaves together biology, behavior, and advanced analytics.

REFERENCES

Kaggle diabetes prediction dataset , retrieved from
<https://www.kaggle.com/datasets/iammustafatz/diabetes-prediction-dataset> .

APPENDIX A - R CODE

R code used for the statistical analysis :
<https://github.com/yaseminelkhider/R-code-used-for-project/blob/main/main%20code>